

Week 01 Task 2

Networking Basics for Defenders

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1 IP ADDRESS

1.1 what is an IP address?

IP stands for Internet Protocol. It is a logical address assigned to every device connected to a network. It will help devices to communicate from one end device to another end device. It is used to communicate over a WAN(Wide Area Network) or LAN(Local Area Network). There are two types of IP address IPV4 and IPV6. It is further divided in Private and Public IP address.

1.2 IPV4

IPV4 (Internet Protocol Version 4) is based on 32 bit address. It has 4 octets. Eachoctet contains 8 bits. The format of IPV4 is 192.168.1.1 It supports 4.3 Billion UniqueAddress. Due to limited addresses subnetting is used.

1.3 IPV6

IPV6 (Internet Protocol Version 6) is based on 128 bit address. It has 8 groups. Eachgroup is based on 16 bits. The format of IPV6 is 2001:0eb8:85a1:0000:8a2e:0370:7334:0380. IPV6 provides much larger address space.

1.4 Difference between IPV4 and IPv6

Feature	IPV4	IPV6
Address Length	32 bit	128 bit
Address Format	Decimal	Hexadecimal
Example Format	192.168.1.x	2001:0eb8:85a1:0000:0370:7334:0380
Security	Basic needs	Built in IPsec Support
Number of Groups/Octets	4 octets	8 groups
Broadcast	Supports BroadCast	Not Supported Broadcast. Uses Multicast

2 Protocols

2.1 TCP

TCP stands for (Transmission control protocol) is a connection oriented protocol that ensures reliability and accuracy of data delivery. It checks for errors, packet delivery that's why it is slow as compare to UDP. It uses 3 way handshake.

2.1.1 UDP

UDP stands for (User Datagram protocol) is a connection less protocol that is simple and fast because it does not ensure reliability of data means some packets maybe lost during transmission.

2.2 HTTP

HTTP (Hypertext Transfer Protocol) is an application-layer protocol used for communication between web clients and web servers. It serves resources such as web pages, images, and API data through request and response messages. HTTP itself does not encrypt transmitted data.

2.3 DNS

DNS (Domain Name System) is responsible for translating human-readable domain names, such as example.com, into IP addresses that computers use to communicate. It typically uses UDP port 53 for standard queries and may use TCP port 53 for larger responses and operations such as zone transfers.

2.4 SMB

SMB (Server Message Block) is a network protocol primarily used for sharing files, folders, printers, and other resources between systems on a network. Modern SMB commonly operates over TCP port 445.

3 OSI Model

3.1 what is the OSI Model?

OSI Stands for Open System Interconnection Model. It's a conceptual Model that explains how data travel from one device to another device with in a network. It contains seven layers each layer has its own responsibility.

3.2 Seven Layers of OSI Model

The seven Layers of OSI Model are listed below

Application Layer: Provides Network services directly to end users applications.

Example: Chrome, Gmail, WhatsApp,HTTP,HTTPS,DNS.

Presentation Layer: This layer provides Compression, Encryption and Decryptionof data for the application layer.

Example: SSL/TLS, JPG, PDF

Session Layer: This layer is use to establishes, synchronizes, and terminates the session between two applications.

Example: Session establishment, management and termination during a video call via zoom, google meet etc.

Transport Layer: This layer provides end to end communication between different devices, flow control and error detection to ensure data moves in a reliable way. It supports two protocols UDP and TCP. Choosing between TCP and UDP completely depends on the situation what it is. Here Data is in the form of segments.

Example: Live video streaming of a cricket match is an example of transport layer that uses UDP protocol because here if some packets of data are lost then there is no issue we can tolerate this

because in Live streams speed matters then perfection.

Network Layer: Receives a segment from Transport layer. Adds an IP Header thencreates a Packet. Chooses the best path to route that Packet using a route table. Routers works here.

Example: Sending email over the Internet where router uses route table to efficientlyroute message to the correct destination.

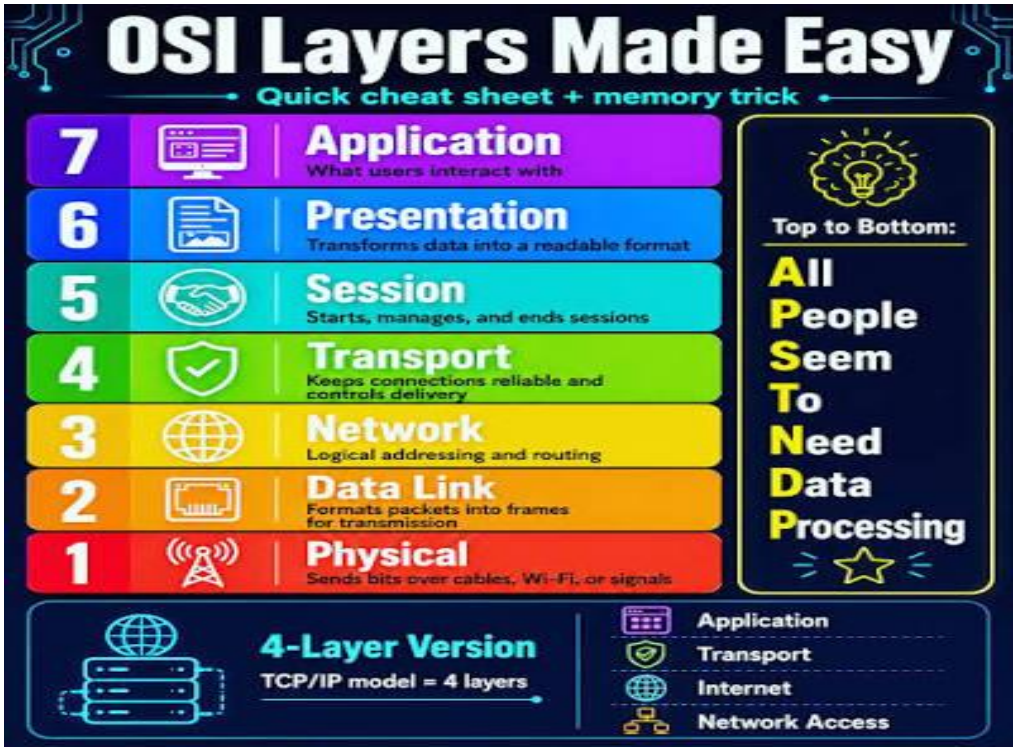
Data Link Layer: Data is received from network layer in Data Link Layer. Data Link Layer creates Frames. It adds a header to a Packet received from Network Layer. Source Mac and Destination Mac. Switches work here.

Example: Sharing a file from one laptop to another over a LAN the data reaches tocorrect destination after identifying the Source MAC and Destination MAC.

Physical Layer: Transmits raw bits over the Physical wire by encoding as electrical or optical signals. For wireless medium it uses radio waves.

Example: An Ethernet cable pulsing electricity or a Wi-Fi router emitting radio waves.

3.3 Pictorial Representation of OSI Model



4 Common Network Ports

4.1 What are Network Ports?

Network Ports are logical communication end point on which network applicationandservices runs. There are total **0-65535** ports in computer Network. The IP address helps to identify the system but Port Number is required to direct data to the actual application or service running within that system. There are 3 major categories of ports

1. 0-1023:--> Well-known Ports/ System Ports
2. 1024-49151:--> Registered Ports/ User Ports

3. 49152-65535:--> Ephemeral Ports

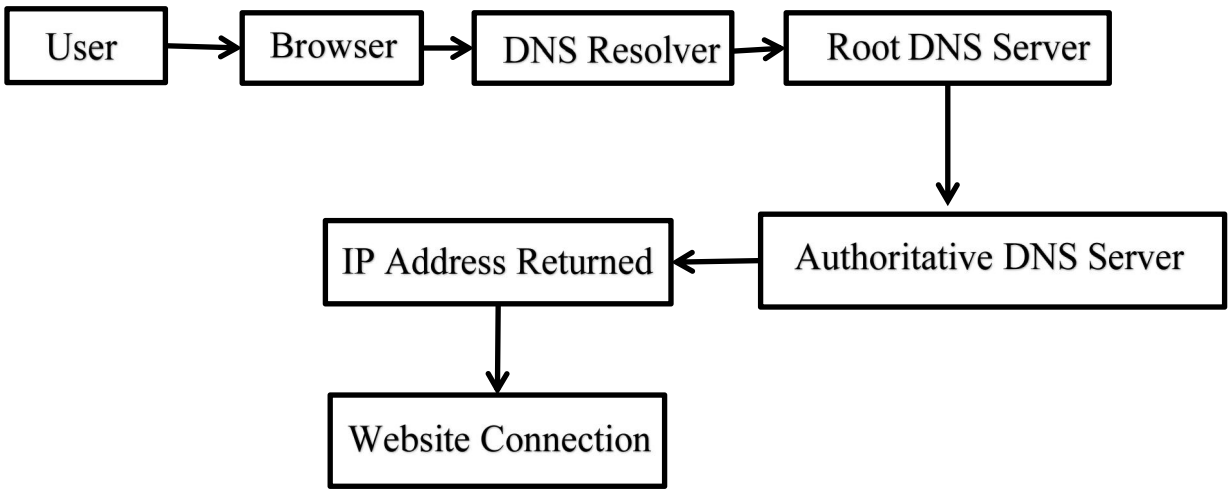
4.2 Common Network Ports

S.No	Port Number	Port Functionality
1	21	Port use for FTP (**File Transfer Protocol**) service to share files between Client and Server
2	22	Port use for SSH(**Secure Shell**) service connection use for secure encrypted remote login on servers
3	25	Port use for SMTP (**Simple Mail Transfer Protocol**) service use to send emails
4	53	Port use for DNS (**Domain Name System**) service use to resolve Domain names like google.com into machine readable IP address .
5	80	Port use for HTTP (**Hyper Text Transfer Protocol**) use for web browsing Here traffic is not encrypted over the web
6	443	Port use for HTTPS (**Hyper Text Transfer Protocol Secure**) use for secure communication over the web communication is encrypted via TLS/SSL.
7	3389	Port use for RDP (**Remote Desktop Protocol**) service use for remotely accessing and controlling another computer's desktop over a network
8	23	Port use for Telnet Provides unencrypted remote access
9	110	Port use for POP3 (**Post Office Protocol v3**) use to receive incoming mails from a server
10	445	Port use for SMB (**Server Message Block**) service use to share files, printers over the network

5 How DNS works Resolution flow

DNS resolution is the process of converting a human-readable domain name, such as **www.example.com**, into an IP address that a computer can use to communicate with the destination server.

DNS Resolution Flow:

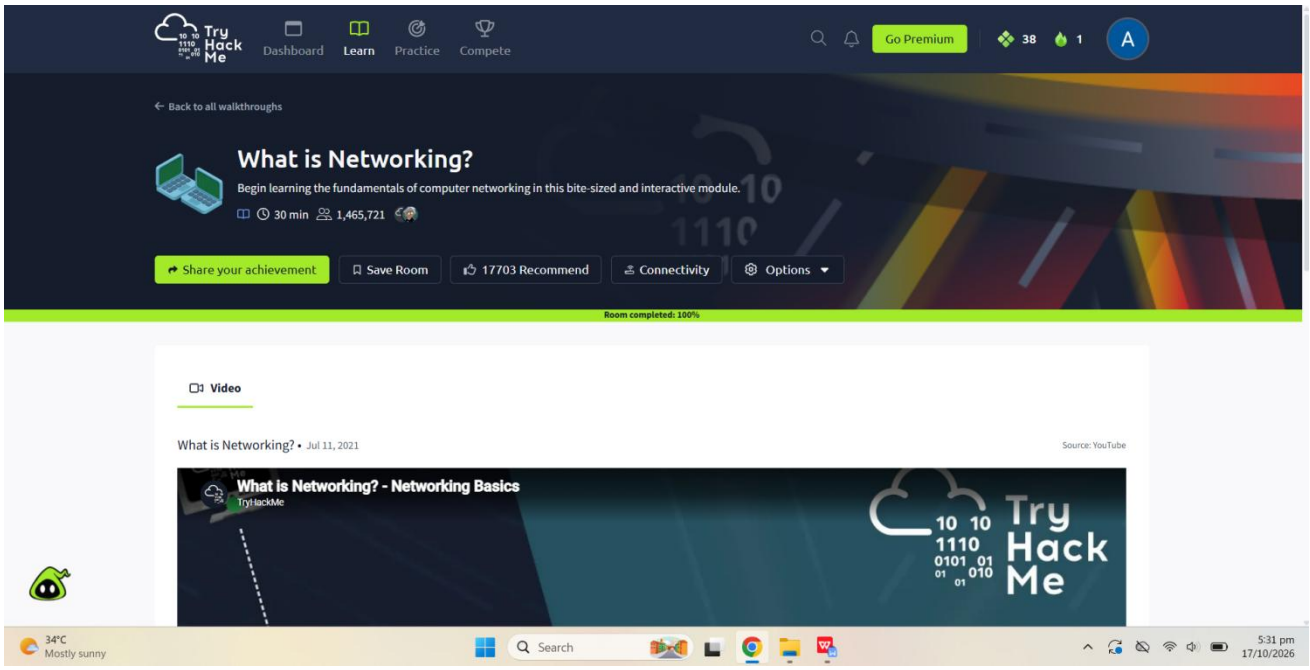


- **User Request:** The user enters a domain name such as `www.example.com` in the browser.
- **Local Cache Check:** The browser and operating system first check whether the domain's IP address is already cached.
- **DNS Resolver:** If no cached record exists, the request is sent to a recursive DNS resolver, usually provided by an ISP or configured DNS provider.
- **Root DNS Server:** The resolver contacts a Root DNS server, which directs it to the appropriate Top-Level Domain (TLD) server.
- **TLD Server:** The TLD server, such as the server responsible for `.com`, directs the resolver to the domain's authoritative DNS server.
- **Authoritative DNS Server:** The authoritative server provides the requested DNS record containing the destination IP address.
- **Response:** The resolver returns the IP address to the user's system and typically caches it according to its TTL.
- **Connection:** The browser uses the returned IP address to connect to the destination web server.

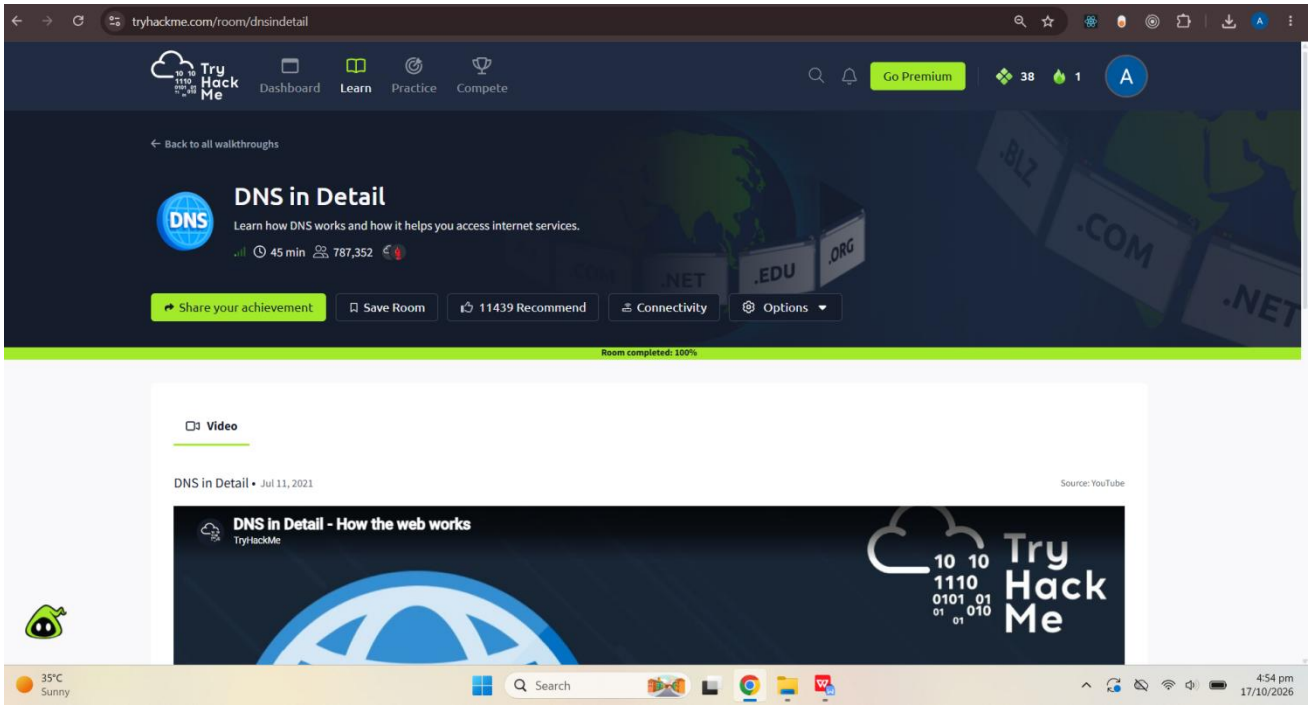
6 Example of Normal Traffic vs Suspicious Traffic

Normal	Suspicious
Regular web browsing using HTTP/HTTPS.	A single IP sending hundreds of requests within a short period.
Authorized user logging in during normal activity.	Multiple failed login attempts from the same IP address.
Traffic using expected ports and protocols.	Unexpected port scanning or connection attempts across multiple ports.

7 TryHackMe What is Networking Room



8 TryHackMe DNS in Detail Room



9 Conclusion

This task strengthened my understanding of overall Networking Concepts as Networking is very crucial to understand before going deep into ethical hacking because attack also happens layer wise like on Data Link Layer , Network Layer etc. Attackers also uses different protocols of networking such as FTP,SSH,SMTP,SMB etc.